CR23532 – CYBER AND DIGITAL FORENSICS

CASE STUDY ASSIGNMENTS – II

B.E. CSE – Cyber Security

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ASSIGNMENT – 2(1)

Case Study: Suspected Unauthorized Access to a College Server

Unique scenario selected for this assignment

A college examination department maintains a restricted server folder containing internal question-bank and assessment documents. One morning, the administrator notices that several files were accessed outside normal working hours and one file appears to have been copied to removable storage. The available evidence consists of the administrator's complaint, a suspect workstation, server access logs, a USB device recovered from that workstation and several relevant files.

Assumptions

- The investigation is formally authorised by the college and is limited to the systems and data relevant to the complaint.
- The workstation, server logs and USB device can be preserved before detailed examination.
- The clocks of different systems may not be perfectly synchronised, so timestamp correlation will be treated carefully.
- The investigation team is trained to handle digital evidence and will maintain a chain of custody.

Question 1 – Explain how the forensic team should move through the Identification, Collection, Examination, Analysis, and Presentation phases for this case.

Digital forensics is normally carried out as a sequence of controlled stages. In this case, the purpose is not simply to find a suspicious file; the team must determine what happened, preserve the evidence, and present a conclusion that another person can independently evaluate.

Identification: The team first identifies the allegation, the systems involved, the suspected time period, relevant users and the types of evidence that may answer the question. The administrator's complaint is recorded as the starting point. At this stage, investigators should not assume that the complaint itself proves unauthorised access.

Collection: The workstation and USB device are secured and their condition is documented. Server logs are preserved from the relevant period. Investigators use appropriate write protection during acquisition where applicable, create forensic copies, calculate hashes and record each transfer. The original evidence is then protected from unnecessary handling.

Examination: Examination is performed on verified forensic copies. Investigators inspect server access records, workstation user/account artifacts, file-system information, removable-device artifacts and the relevant examination files. The objective is to extract useful artifacts without changing the original evidence.

Analysis: The extracted artifacts are correlated. For example, a server access at a particular time can be compared with a workstation logon, access to the same file, a USB connection and evidence of a copy. The analyst must also consider alternative explanations such as automated jobs, shared accounts or incorrect system clocks.

Presentation: The final stage converts the technical examination into understandable findings. The report should show what was observed, how it was obtained, what it means, what remains uncertain and why the conclusion is reasonable.

Question 2 – Prepare a forensic investigation flowchart showing the sequence of activities from the initial complaint to presentation of findings.

The following flowchart shows the complete investigation sequence for this particular case. Each stage should be documented before moving to the next major stage.

1. Administrator reports suspected unauthorised access
   ↓
2. Define allegation, scope, time period and authority
   ↓
3. Identify workstation, server logs, USB and relevant files
   ↓
4. Secure evidence and document initial condition
   ↓
5. Acquire forensic copies using appropriate protection
   ↓
6. Calculate and record integrity hashes
   ↓
7. Examine copies and extract relevant artifacts
   ↓
8. Correlate accounts, files, devices and timestamps
   ↓
9. Build and validate the event timeline
   ↓
10. Consider alternative explanations and limitations
   ↓
11. Prepare findings, exhibits and final forensic report
   ↓
12. Present conclusion to technical and non-technical authorities

Figure 1. Investigation flowchart for the suspected examination-server access.

Question 3 – Identify at least five items of potential digital evidence in the case and classify the evidence according to its investigative value.

Different evidence items answer different questions. Some provide direct technical records of an event, while others provide background or corroboration. The value of an item should therefore be described in relation to the question being investigated.

Evidence item	Classification	Investigative value / possible use

Evidence item	Classification	Investigative value / possible use
Server access logs	Primary technical evidence	Can show account, source host, requested resource, access result and time recorded by the server.
Suspect workstation	Primary digital evidence source	May contain user profiles, recent activity, file metadata, application artifacts and removable-device traces.
Recovered USB device	Primary removable-media evidence	May contain copied files, metadata, file-system records and device-identification information.
Relevant examination files	Potential direct content evidence	Can show the actual documents involved and allow comparison with copies found elsewhere.
Administrator's complaint	Contextual / initiating evidence	Defines when the problem was noticed and what was initially reported; it is not by itself proof of the incident.
User/account artifacts	Corroborating evidence	Can connect a logical account or session with activity occurring on the workstation.
File-system metadata	Corroborating technical evidence	Provides filenames, paths, sizes, attributes and timestamps for comparison with other sources.

Question 4 – Explain the precautions required during collection and preservation so that the evidence remains reliable and defensible.

- Obtain proper authority before collecting or examining the device. The scope should be clear so unrelated information is not unnecessarily examined.
- Photograph or document the device condition, labels, cable connections and visible screen state before handling.
- Record whether the workstation is powered on or off. Do not casually browse a live system because normal interaction can change evidence.
- Use suitable write-blocking methods when acquiring storage media so that the source is not unintentionally modified.
- Create a forensic image or other approved forensic copy and perform the normal examination on the copy rather than repeatedly handling the original.
- Record acquisition tool/version, source identifier, destination, date/time, examiner, settings, errors and hash values.
- Seal and store the original evidence in a secure, access-controlled location.
- Maintain a chain-of-custody record for every movement or transfer.
- Preserve confidentiality because examination-related files may contain sensitive academic information.
- Document limitations rather than silently changing or correcting evidence. For example, if a system clock is wrong, record the offset and explain how it affects the timeline.

Question 5 – Describe how investigators can establish a timeline of the suspected unauthorized access using available evidence.

A forensic timeline combines events from different sources and places them into a common chronological framework.

The investigator should first preserve the original timestamp values and then determine the relevant time zone and clock settings before making comparisons.

The server log may provide the strongest starting point because it records requests made to the server. The investigator can then compare the server time with workstation logon records, recent-file/application artifacts and USB connection activity. If the same account, host, file path and time window appear across multiple sources, the event sequence becomes more strongly supported.

The timeline should not treat a timestamp as proof of a person's physical action. A file timestamp indicates when an artifact was recorded or changed according to that system; it does not automatically identify the person who caused the event.

Illustrative time	Source	Observed event	Interpretation
21:41:12	Server log	Restricted folder requested by a known account	Shows a server-side access event.
21:42:03	Workstation	Relevant file appears in recent-access artifact	Supports local interaction with the file.
21:47:19	USB artifacts	Removable device activity is recorded	Shows the USB was active around the period.
21:48–21:53	File/USB metadata	Matching file name or copy-related traces appear	Potential evidence of transfer; needs corroboration.
22:06:31	Server log	Session/request activity ends	Provides an approximate end boundary.

Question 6 – Explain what should be included in the final forensic report and how findings should be presented to a non-technical authority.

A forensic report should be detailed enough for a technical reviewer to understand the method but simple enough for a decision-maker to understand the conclusion. The report should not contain unexplained technical terms when a plain-language explanation can be given.

- Case identification, investigator details, date and authority for the examination.
- Purpose and scope of the investigation.
- Evidence inventory and condition at the time of receipt.
- Chain-of-custody summary.
- Acquisition method, tool/version, image identifiers and hash values.
- Examination methods and important artifacts examined.
- Chronological timeline and evidence-correlation table.
- Findings stated as observations followed by their interpretation.
- Alternative explanations and limitations.
- Final conclusion answering the original investigative question.
- Appendices containing relevant screenshots, logs, hash lists or artifact extracts where permitted.

For a non-technical authority, the investigator should begin with a short executive summary. For example: “The available records show that the restricted examination file was accessed from the identified workstation during the investigated period, and removable-media activity occurred shortly afterward. These records are consistent with a possible file-copying event. However, the evidence does not by itself establish which individual physically operated the workstation.”

Question 7 – Briefly discuss the legal and ethical considerations involved in collecting digital evidence in a college environment.

- The investigation must have appropriate institutional and legal authority. Investigators should not search devices merely because a person is suspected.
- Only information relevant to the authorised investigation should be collected or examined where possible.
- Student and staff information must be kept confidential and accessed only by authorised personnel.
- Investigators should remain impartial and record evidence that supports as well as contradicts the working hypothesis.
- Evidence must be preserved using repeatable and defensible procedures so that another examiner can review the work.
- Investigators should not manufacture certainty. If attribution is weak or evidence is incomplete, that limitation must appear in the report.
- Retention, disclosure and access to evidence should follow the institution's applicable policies and legal requirements.

ASSIGNMENT – 2(2)

Case Study: Linking a Suspicious File to a User Using File-System Evidence

Unique scenario selected for this assignment

A desktop computer is shared by several staff members in a design office. During a routine security check, investigators find a suspicious executable named “sync_patch.exe”, hidden files, encrypted documents and browser artifacts.

Several user accounts exist on the computer. The investigation focuses on determining whether file-system and application artifacts can reasonably associate the suspicious file with one user account.

Assumptions

- The investigation team has an authorised forensic image or other approved evidence source.
- The suspicious filename is only an investigative lead; the file is not automatically considered malicious.
- ILookIX is discussed as the forensic analysis tool named in the assignment. The procedure describes the intended workflow and does not claim that the software was actually executed.

Question 1 – Identify the hardware and software components that should be documented before analysis.

Before analysing a computer, the investigator should document the environment because hardware and software characteristics influence how artifacts are interpreted.

- Computer make, model, asset number and serial number.
- Processor, RAM and storage-device details where relevant.
- Internal disks/SSDs, partitions and storage capacities.
- Connected USB devices and other removable storage.
- Network interfaces and relevant connection state.
- Operating-system name, edition, version and build.
- File-system types used by the storage volumes.
- Installed browsers, email clients, messaging applications and security software.
- User accounts and account types.
- System time zone and clock configuration.
- Encryption, virtualisation, backup or synchronisation software.
- Forensic acquisition and analysis tools, including version numbers.

Question 2 – Explain how storage devices, operating systems, file systems, and metadata can help investigators locate relevant evidence.

Storage devices determine where persistent evidence can exist. A physical disk may contain multiple partitions or volumes, and removable storage may provide another source of relevant information.

The operating system produces artifacts related to user accounts, application execution, configuration, device connections and system events. These artifacts can help establish what environment existed at a particular time.

The file system provides directory structures, file names, allocation information and metadata. Metadata can include file size, path, attributes and several types of timestamps. These properties can help locate suspicious files and compare their activity with user sessions.

For example, if the suspicious executable is found inside one user's profile and a browser artifact shows that the same profile downloaded the file shortly before an execution-related artifact, the relationship becomes stronger. However, the investigator must still consider account sharing, remote access and automatic processes.

Question 3 – Describe how hidden files and encrypted files may affect a forensic investigation.

Hidden files may not be visible through an ordinary graphical file browser, but they remain part of the storage structure and may contain important evidence. A forensic examiner should therefore enumerate file-system entries rather than relying only on what a normal user interface displays.

Encrypted files create a different problem because the content may not be readable even though the file and its metadata can still be observed. Investigators should preserve the encrypted original, document its location and properties, and use only authorised decryption or credential-recovery methods.

Investigators should never alter the original merely to test a password or encryption method. If the content cannot be decrypted, the final report should clearly state that the encrypted material was identified but its contents could not be verified.

Question 4 – Prepare an evidence-linkage table showing how file names, paths, timestamps, user accounts, metadata, and other artifacts could be correlated to a user.

Artifact	Hypothetical observation	Possible relationship to user	Strength / caution
sync_patch.exe	Located at C:\Users\Arun\AppData\Local\Temp	File is inside Arun's profile	Moderate; profile access is not identical to physical-person attribution.
Executable metadata	Timestamp overlaps Arun's logged-on session	Temporal association	Moderate.
Recent-file/application artifact	Program reference appears under Arun's profile	Supports account-level use	Moderate to strong if independently corroborated.
Browser download artifact	File downloaded to same profile shortly before activity	Links file to browser/profile	Strong corroboration.
Account logon record	Arun account active during the relevant period	Connects time window to account	Moderate.
USB artifact	Removable device connected during same period	Contextual support	Weak to moderate; does not identify operator.

Question 5 – Explain a suitable workflow for importing/acquiring the forensic evidence and analyzing it using ILookIX, based on the concepts covered in the syllabus.

The ILookIX workflow should be understood as an examination process performed after evidence has been properly acquired and preserved. The original source should remain protected while the examiner works on the forensic evidence set.

1. Confirm authorisation and evidence inventory
   ↓
2. Use the approved forensic image/evidence copy
   ↓
3. Import/register the evidence in ILookIX
   ↓
4. Record case details and verify evidence identity/integrity
   ↓
5. Enumerate volumes, file systems and user profiles
   ↓
6. Search for suspicious filenames, paths and extensions
   ↓
7. Examine file metadata and application/user artifacts
   ↓
8. Correlate timestamps, account activity and browser records
   ↓
9. Tag relevant artifacts and record observations
   ↓
10. Validate important findings against underlying evidence
   ↓
11. Prepare screenshots/exports and document the conclusion

The important point is that the tool helps organise and examine evidence; the final conclusion must come from the underlying artifacts and their correlation, not from a tool-generated label alone.

Question 6 – Differentiate between an artifact that directly links a user to an activity and an artifact that merely provides supporting/contextual information.

Type	Example	Explanation
Direct or stronger account linkage	An authenticated user session contains an artifact generated during that same session	Provides a stronger technical association between the account and the event, although account compromise/shared use must still be considered.
Contextual artifact	USB device connection around the same time	Shows that a device was present but does not identify the operator.
Contextual artifact	File creation timestamp	Shows when the file system recorded creation; it does not automatically identify the person.

Type	Example	Explanation
Stronger corroboration	User-specific browser record shows the suspicious file was downloaded to the same profile	Connects an account/profile, application and file in a more meaningful way.

Question 7 – Identify at least three limitations or alternative explanations that investigators should consider before attributing the activity to a particular user.

- The user account may be shared, delegated or compromised. Account attribution is not automatically physical-person attribution.
- The system clock may be inaccurate, or timestamps may be stored in different time zones.
- Applications and operating-system services can generate files automatically.
- Another user may have had access to the profile or the workstation while the account remained logged in.
- Remote administration or remote desktop activity could make a local artifact appear to be a local keyboard action.
- Synchronisation, backup or security software may create or modify files without deliberate user interaction.
- Forensic parsers may not correctly interpret every application version or unsupported artifact.

Conclusion: A defensible attribution requires multiple independent artifacts that converge on the same account, time and activity. A filename or profile path alone is not sufficient.

ASSIGNMENT – 2(3)

Case Study: Forensic Imaging and Safekeeping of a Suspect Storage Device

Unique scenario selected for this assignment

A laptop is seized during an investigation into suspected theft of proprietary project data. An attached USB drive is also collected because investigators believe files may have been copied to it. The objective is to preserve both devices, create reliable forensic images, verify their integrity and retain the evidence safely for later examination.

Assumptions: the seizure is authorised; the devices can be physically isolated; the laboratory has an approved imaging workstation, suitable write protection and a validated workflow involving IXImager and the ASB container format.

Question 1 – Prepare a step-by-step procedure for safely collecting and transporting the laptop and USB device before imaging.

1. Confirm legal/institutional authority and identify the devices

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2. Photograph labels, ports, cables and visible condition

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3. Record whether the laptop is powered on or off

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4. Record any visible screen information without unnecessary interaction

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5. Avoid unnecessary network or user interaction

↓

6. Package laptop and USB separately in labelled evidence containers

↓

7. Record serial numbers/identifiers and initial condition

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8. Seal the packages and begin the chain-of-custody record

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9. Transport using controlled and access-limited handling

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10. Place evidence in secure storage until laboratory acquisition

If the laptop is powered on, trained investigators must follow the laboratory's approved live-response procedure if volatile evidence is considered important. The investigator should not randomly click through the system because this can change evidence.

Question 2 – Explain the purpose of physical acquisition and why analysis should normally be performed on a forensic image rather than the original storage media.

Physical acquisition attempts to obtain a low-level representation of the storage device, including allocated and potentially unallocated areas. This is important because a normal file copy may miss deleted remnants, file-system structures and other low-level evidence.

The original storage device should normally be preserved as evidence while examination is performed on a forensic image. Opening files, booting the operating system or mounting the original can change timestamps, logs, caches or allocation information. Working on a verified image reduces this risk and also allows another examiner to repeat the analysis without changing the original.

Question 3 – Design a forensic imaging workflow using IXImager. Include source identification, write protection, imaging, verification, hashing, and documentation.

1. Identify the source device and record serial number/capacity

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2. Connect the source through appropriate write protection

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3. Select the approved acquisition method and destination

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4. Enter case, evidence and examiner identifiers

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5. Acquire the device into the approved forensic image/container

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6. Record acquisition progress and any errors

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7. Calculate cryptographic hash values

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8. Verify the acquired image against the source

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9. Secure the original device after successful verification

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10. Use only the verified image for normal examination

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11. Record tool version, settings, hashes and verification result

The procedure is defensible because every important transition is recorded: what was acquired, how it was acquired, whether the source was protected, whether the image matched the source and where the original was stored.

Question 4 – Explain how hash values can be used to demonstrate integrity before and after acquisition.

A cryptographic hash is a fixed-length value calculated from the contents of data. When the same data is hashed using the same algorithm, the expected hash value should be the same.

During forensic acquisition, the investigator can record the hash of the source and the corresponding acquired image or otherwise perform an image-verification operation. A matching value provides evidence that the acquired representation corresponds to the source data at the time of acquisition.

Hashing proves integrity of the data representation; it does not prove who created the data, whether the file is malicious or whether the allegation is true. Therefore hash values should be combined with chain-of-custody and acquisition records.

Question 5 – Describe the role of an ASB container in storing or managing forensic evidence and identify the information that should accompany the evidence.

An ASB container can be used as a structured forensic evidence container where the laboratory workflow supports it.

Its purpose is to keep evidence data and relevant descriptive information together in a controlled format. The exact fields and implementation depend on the laboratory/tool environment, so the examiner should document the format/version used rather than assuming every ASB implementation behaves identically.

- Case identifier and evidence number.
- Source-device identifier and serial number.
- Acquisition date and time.
- Examiner/operator identity.
- Acquisition method and tool/version.
- Hash algorithm and recorded hash value(s).
- Verification result.
- Source condition and any acquisition errors.
- Chain-of-custody information.
- Relevant notes about write protection, storage location and subsequent handling.

Question 6 – Explain data carving and discuss when carving may be necessary if file-system information is unavailable or damaged.

Data carving is a recovery technique that searches raw storage for known file signatures or structural patterns. It can reconstruct files even when normal directory information is missing or damaged.

Carving is useful when a file has been deleted, a file system is corrupted, or directory metadata cannot be trusted. For example, an image file may still contain a recognisable header and internal structure even after its directory entry has disappeared.

However, carving is not perfect. Fragmented files may be reconstructed incorrectly, signatures may create false positives, and the recovered file may be incomplete. Therefore carved data must be

validated using file structure, surrounding storage information, hashes where possible and independent artifacts.

Question 7 – Discuss date/time problems in digital evidence and explain how investigators should document time-zone, clock, and timestamp uncertainties.

- Record the device's configured clock and time zone where available.
- Preserve the original timestamp exactly as observed before applying any conversion.
- Distinguish UTC from local time and document every conversion.
- Compare the device clock with independent records when possible.
- Consider clock drift, manual clock changes, virtual machines and application-specific timestamp behaviour.
- Do not present a normalised time as if it were the original evidence value.
- When uncertainty exists, report an approximate time range and explain the reason for the uncertainty.

Question 8 – Prepare a chain-of-custody table for the seized laptop from collection through laboratory examination.

Stage	Date/time	Custodian	Action and condition	Record
Collection	04-Oct-2026 10:15	Field examiner	Laptop and USB identified, photographed and sealed	Collection entry
Transport	04-Oct-2026 10:40	Evidence courier	Sealed package received intact	Transfer entry
Storage	04-Oct-2026 11:05	Evidence custodian	Placed in restricted evidence locker	Storage entry
Lab receipt	05-Oct-2026 09:00	Lab examiner	Seal checked; package condition recorded	Lab receipt
Imaging	05-Oct-2026 09:20	Lab examiner	Source connected through write protection and imaged	Acquisition log
Verification	05-Oct-2026 11:30	Lab examiner	Image hash/verification recorded	Verification record
Examination	05-Oct-2026 14:00	Forensic analyst	Verified image examined; original remains secured	Examination log

ASSIGNMENT – 2(4)

Case Study: Recovery of Deleted Evidence from Volume Shadow Copies

Unique scenario selected for this assignment

An employee is suspected of deleting documents and browser-related artifacts after an unauthorised activity. A forensic image of the Windows system is available, and investigators discover that Volume Shadow Copies are present. The investigation therefore focuses on selecting relevant evidence, analysing historical system states and validating recovered artifacts before using them in the final report.

Assumptions: the forensic image has already been acquired lawfully and its integrity has been verified. The presence of a Volume Shadow Copy does not guarantee that every deleted artifact will be recoverable.

Question 1 – Explain how investigators should select and categorize digital evidence before detailed analysis.

Evidence selection should begin with the investigative question rather than with random searching. The team should define what event is being investigated, the likely time period, the systems involved and the types of artifacts that could answer the question.

After inventorying the available evidence, investigators can classify artifacts according to their role. Relevant artifacts should be prioritised, but potentially unrelated material should not be discarded without documenting why it was considered outside scope. Every important artifact should retain a traceable source identifier.

Question 2 – Prepare a classification scheme for user-created files, system/application files, potentially suspicious files, deleted/recovered artifacts, and supporting/contextual evidence.

Category	Examples	Purpose
User-created files	Documents, spreadsheets, images, notes	May contain the actual content involved in the incident.
System/application files	Event logs, registry-related artifacts, application databases	Record system and application activity.
Potentially suspicious files	Unexpected executables, scripts, unusual archives	Identify objects requiring deeper examination; suspicious does not mean proven malicious.
Deleted/recovered artifacts	Recovered documents, browser records, file remnants	Can show historical activity or content no longer visible in the active file system.
Supporting/contextual evidence	User logons, system configuration, network records, clock settings	Helps explain and corroborate the primary artifacts.

Question 3 – Explain how Event Analysis can assist in reconstructing a sequence of activities.

Event Analysis brings artifacts from different sources into a chronological sequence. Instead of examining each artifact separately, investigators compare events such as logons, application execution, file access, browser activity and deletion records.

For example, a user logon followed by access to a document, browser activity and later deletion may form a technically consistent sequence. If a Volume Shadow Copy contains an earlier version of the document, it can provide historical evidence that the document existed before the later deletion.

The investigator must still distinguish observation from inference. A timeline can show that several events occurred close together; it does not automatically prove that one person intentionally caused every event.

Question 4 – Explain the purpose of Cloud Analysis and Lead Analysis in an investigation and identify the kinds of questions each may help answer.

Cloud Analysis: Cloud Analysis examines authorised cloud-service information such as account activity, synchronisation records, cloud files, access records or provider logs. It may answer questions such as whether a file was synchronised to a cloud service, when an account accessed a service, or whether a local file has a cloud counterpart.

Lead Analysis: Lead Analysis uses one observation to decide what should be examined next. For example, discovering a suspicious filename can lead to searches for the same name on other volumes, accounts, email messages or cloud storage. A lead directs the investigation but is not itself a final finding.

Question 5 – Describe how Volume Shadow Copy analysis can help recover or compare earlier versions of evidence.

Volume Shadow Copy (VSS) can preserve earlier states of files or system information. If an investigator later finds that a document was deleted or changed, an earlier snapshot may still contain the previous version.

The examiner should identify the available snapshots, record their timestamps, access them through a controlled forensic workflow, locate the relevant path and compare the historical artifact with the current image. The recovered artifact should then be extracted, hashed and correlated with other evidence.

VSS is therefore particularly useful for historical comparison. It may show that a file existed previously even though the active file system no longer contains it.

1. Verify forensic image and identify available VSS snapshots

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2. Record snapshot identifiers and timestamps

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3. Access the selected snapshot using a controlled forensic method

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4. Locate the relevant volume, path and artifact

↓

5. Compare historical state with current state

↓

6. Extract and preserve the relevant artifact

↓

7. Hash and document the recovered object

↓

8. Correlate with logs, user activity and other artifacts

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9. Validate interpretation before reporting

Figure 2. Volume Shadow Copy recovery workflow.

Question 6 – Construct a hypothetical timeline showing how a file could be created, modified, copied, deleted, and later recovered from a Volume Shadow Copy.

Time	Event	Evidence source	Meaning
09:12	project_notes.docx created	File-system metadata	The file existed in the active system.
10:05	Document modified	Metadata/application artifact	The contents or metadata changed.
10:32	Document copied to another folder	File-system/application evidence	A second location may have been created.
11:10	Active copy deleted	Deletion/file-system evidence	The current path no longer contains the copy.
12:00	VSS snapshot represents earlier state	VSS metadata	Historical system state is available.
13:20	Earlier document recovered from VSS	Forensic examination record	Historical version is available for validation.

Question 7 – Explain how the recovered artifact should be validated before it is included as a finding in the final report.

- Confirm the identity of the VSS snapshot from which the artifact was recovered.
- Record the snapshot timestamp and original path.
- Document the extraction method and examiner.
- Calculate a hash for the recovered object.
- Check whether the recovered file is structurally complete or corrupted.
- Compare the recovered artifact with current-state evidence and other independent sources.
- Determine whether it is a normal recovered file or a reconstruction produced by carving/parsing.
- Clearly distinguish observed facts from the investigator's interpretation.
- State any limitations that could affect confidence in the recovery.

Question 8 – Identify possible sources of error or alternative explanations that could affect the interpretation of timestamps and deleted artifacts.

- System clocks may be inaccurate or may have been manually changed.
- Different artifacts may represent different kinds of time, such as creation time, modification time, access time or log-write time.
- Time-zone conversion may create an apparent difference between sources.
- VSS snapshots are created according to system processes and do not necessarily occur exactly when an alleged action happened.
- Automated applications, synchronisation tools or backup processes can modify files without direct user interaction.
- Deletion from the active file system does not necessarily mean secure destruction.
- Parser limitations or damaged evidence may produce incomplete results.
- A recovered file's existence does not by itself prove who deleted the original.

ASSIGNMENT – 2(5)

Case Study: Multi-Source Investigation of a Suspected Cyber Incident

Unique scenario selected for this assignment

A student account is suspected of being involved in unauthorised access to an online service. Investigators, subject to proper authorisation, have access to a Windows computer, browser artifacts, email messages, messaging records, a mobile phone and relevant Internet/cloud records. Since the evidence is distributed across several sources, the main challenge is to determine whether the artifacts form one consistent sequence of activity.

Assumptions: all sources are examined only under proper authority. Provider-side records may be incomplete because of retention policies. The scenario is hypothetical and does not accuse any real student.

Question 1 – Identify the potential evidence available from Windows and other operating systems in this case.

Source	Potential evidence
Windows computer	Event logs, account/logon records, file-system metadata, browser databases, registry/configuration artifacts, recent-file records, installed software and removable-device traces.
Linux/macOS or another desktop OS	System/application logs, user profiles, file-system metadata, browser artifacts, shell/command history where available and authorised, and device-connection records.
Mobile operating system	Application databases, messages/calls, account information, browser records, media metadata and other application-specific artifacts where lawfully available.
Cloud/online services	Authentication logs, access times, source network information where retained, cloud-file activity, synchronisation records and service-security events.

Question 2 – Explain what browser artifacts may reveal about user activity and why browser evidence must be interpreted carefully.

Browser artifacts may include browsing history, search queries, downloaded files, cookies, cache records, bookmarks, session-related information and browser-profile data. These artifacts can help establish that a browser profile recorded a particular web address or downloaded a particular file during a particular period.

Browser evidence must be interpreted carefully because it does not automatically prove human intention. Pages can be opened by redirects, extensions, applications or synchronisation. History can also be deleted or synchronised between devices. Private browsing may reduce the amount of stored evidence. Therefore a browser record is much stronger when it agrees with server logs, local files, email records or other independent sources.

Question 3 – List the types of evidence that may be available from email and messaging systems and explain how they can be correlated with other artifacts.

- Email headers and metadata, including sender/recipient information, message identifiers, routing information and timestamps.
- Message bodies and attachments that may describe activity or contain relevant files/links.
- Account login and security notifications, where retained and lawfully obtainable.
- Messaging timestamps, participants, attachments and service metadata.
- Correlation with browser downloads can show that an attachment or URL mentioned in a message was later accessed.
- Correlation with local file hashes/names can help determine whether an attachment corresponds to a file found on a computer or mobile device.
- Correlation with cloud records can help establish whether a referenced file was synchronised or accessed online.
- The content of a message should be treated as a source of information, not automatically as proof that the described event actually occurred.

Question 4 – Explain the forensic considerations when examining a mobile phone and identify challenges associated with mobile evidence.

The examiner should identify the device model, operating-system version, encryption/lock state, SIM/eSIM context where relevant, acquisition method and available lawful extraction options. The device should be preserved using the laboratory's approved procedure because network synchronisation or normal application activity can change data.

Mobile evidence can be difficult because devices may be encrypted, locked or protected by secure hardware.

Applications frequently store information in proprietary databases, may delete or expire messages, and may keep important data in the cloud rather than on the phone. A single account can also be active on multiple devices.

The investigator should therefore distinguish device-resident evidence from cloud-resident evidence and document exactly what extraction method was used and what was unavailable.

Question 5 – Discuss Internet and cloud-based evidence sources that may be relevant to the investigation.

- Online-service authentication and account-security logs.
- Cloud-storage file activity, version history and synchronisation records.
- Provider-side access records and network information where retained and lawfully obtainable.
- Institutional DNS, proxy, firewall or network logs where available and within scope.
- Cloud email and messaging records.
- Account recovery events, security alerts and multi-factor authentication records.
- Publicly available service information or incident records when relevant.

Question 6 – Prepare a cross-source correlation table linking at least five hypothetical artifacts from the computer, browser, email/messaging, mobile, and cloud sources.

Time	Computer	Browser / email	Mobile	Cloud / Internet	Correlation
08:14	Windows account logon	Browser session begins	Security notification received	Cloud authentication record	Strong temporal convergence; verify device/account identity.
08:21	Downloaded file appears in profile	Email attachment download recorded	Message references related file	Cloud file access recorded	Possible transfer chain; compare file properties and hashes.
08:29	Browser requests target service	History records target URL	No matching app event	Service records request	Supports laptop as possible source if network evidence agrees.
08:36	Local file opened	Browser shows related page	Message discusses the file	Cloud copy metadata changes	Multiple-source corroboration; attribution still requires care.
08:52	Session closes	Browser activity stops	Phone remains idle	Cloud authentication ends	Useful boundary for the reconstructed sequence.

Question 7 – Explain at least six major challenges in digital forensics when evidence is distributed across multiple devices and cloud services.

- **Different formats:** Windows, mobile apps, browsers and cloud services store evidence in different structures.
- **Time synchronisation:** device clocks and provider timestamps may use different time zones or have different clock accuracy.
- **Cloud volatility:** provider data may be deleted or become unavailable because of retention policies.
- **Encryption and access control:** devices and cloud accounts may restrict lawful extraction or require authorised credentials.
- **Account sharing or compromise:** an account identifier cannot automatically be equated with one physical person.
- **Data duplication:** synchronisation can place the same artifact on several devices, so apparently independent copies may actually have one common origin.
- **Privacy and scope:** multi-device examination can expose unrelated personal data, so collection must remain within authorised boundaries.
- **Tool limitations:** forensic software may not support every application version or proprietary database.
- **Large evidence volume:** filtering is necessary, but excessive filtering can accidentally remove useful evidence.

Question 8 – Prepare a final investigation conclusion distinguishing direct evidence, corroborating evidence, and unresolved questions.

Direct evidence: These are artifacts that directly record a technical event. Examples include an authenticated cloud-service record showing a login, a preserved message containing a specific

attachment, or a server record showing a request to the online service. Even direct technical evidence must be assessed for source reliability and account attribution.

Corroborating evidence: These are independent artifacts that support the same event sequence. For example, a browser download, matching local file, mobile notification and cloud record that all occur in a compatible time window can make the reconstruction stronger.

Unresolved questions: The investigation may still be unable to prove who physically controlled the account, whether a device was remotely accessed, whether a timestamp was recorded in UTC or local time, whether synchronisation generated some artifacts, or whether missing provider records would change the sequence.

Final conclusion: The most defensible conclusion is therefore a reconstructed sequence supported by multiple sources rather than an accusation based on one artifact. The report should clearly identify what is established, what is corroborated, what remains uncertain and what additional evidence would resolve the uncertainty.

Multi-source evidence map: Windows workstation ↔ browser/email ↔ mobile phone ↔ cloud/service records ↔ institutional network records. Every arrow represents a correlation that must be supported by an identifiable artifact, identifier, timestamp or content relationship.